

MATH 231 – Applied Linear Algebra

Tentative Referenced Textbooks

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The four texts below anchor the course: an applied spine (Boyd & Vandenberghe), a numerical-methods reference (Trefethen & Bau), a geometry/intuition source (Strang), and a theoretical backbone (Axler). Bibliographic details should be verified against the exact printing adopted.

[1] Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares

Author(s): Stephen Boyd and Lieven Vandenberghe
Publisher: Cambridge University Press
Published: 2018
ISBN: 978-1316518960 (hardback)
Role in course: Applied spine (vectors, norms, least squares, applications)
Link: <https://web.stanford.edu/~boyd/vmls/vmls.pdf>

[2] Numerical Linear Algebra

Author(s): Lloyd N. Trefethen and David Bau III
Publisher: Society for Industrial and Applied Mathematics (SIAM)
Published: 1997
ISBN: 978-0898713619 (paperback)
Role in course: Numerical linear algebra (conditioning, stability, QR, eigenvalues, SVD)
Link: <https://www.stat.uchicago.edu/~lekheng/courses/309/books/Trefethen-Bau.pdf>

[3] Linear Algebra and Its Applications (4th edition)

Author(s): Gilbert Strang
Publisher: Cengage Learning (Thomson Brooks/Cole)
Published: 2006
ISBN: 978-0030105678
Role in course: Subspaces and geometric intuition
Link: https://rksmvv.ac.in/wp-content/uploads/2021/04/Gilbert_Strang_Linear_Algebra_and_Its_Applicatio_230928_225121.pdf

[4] Linear Algebra Done Right (4th edition)

Author(s): Sheldon Axler
Publisher: Springer (Undergraduate Texts in Mathematics)
Published: 2024
ISBN: 978-3031410253
Role in course: Theoretical foundation (vector spaces, linear maps, spectral theory)
Link: <https://linear.axler.net>

The PDF links above point to author- or course-hosted copies referenced during course design; confirm access rights and edition alignment before distributing to students.